

EDA Becoming INTERACTIVE With 3D RENDERING And ROUTING



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Electronic design automation (EDA) is an indispensable tool for optimised circuits. Most engineers have worked on one such tool or the other. Marco-Casale Rossi, product marketing manager, design group, Synopsys Inc., explains the advances in EDA, "Today, we can floor-plan hundreds of millions of library instances, or we can place and route tens of millions of library instances in less than 24 hours."

"We have moved from designing 350nm to 10nm," says Ruchir Dixit, technical director, Mentor Graphics. "There was a time when a four-layered board was among the most complex circuit designs. This has become the basic now." We also have changes in the type of components being used and signal speeds, hence calling for efficient designs.

The ever increasing tools

Since its emergence in the late 1970s, EDA has quickly evolved into a dedicated industry, and new players keep emerging at regular intervals. Today, we have advanced EDA tools like Altium Design-

er, LabVIEW and Virtuoso, among others, that help engineers come up with the best blueprint for complex circuits.

Besides the industry-established tools, we also have some Cloud based tools that have revolutionised the way beginners and hobbyists take up electronics. Cloud based EDA tools like EasyEDA and MeowCAD allow them the ease and freedom to work with the latest libraries, while providing updated versions of the software, without the hassle of updating user software.

Cloud based tools. Imagine working on a tablet; you do not have to carry your heavy laptop everywhere. You do not have to think about storage, appropriate graphics and processor requirements. All you need is a browser and you are good to go.

Along with traditional circuit simulation features, Cloud based tools provide ubiquitous access. This allows you to travel light with only the concern for the Internet speed (which is nothing new). Using various advances in website designing, these are now accessible over a wide variety of mobile platforms including smartphones and tablets.

Cloud based tools also feature quick help via dedicated forums, which provide access to the community of like-minded individuals working on their own problems. Sharing your project with the developer community and your co-workers over the Cloud, without having to care for compatibility of software, has also come as an advantage with Cloud based EDA tools.

Flip side to the Cloud. However, Cloud based tools are yet to establish themselves for high-precision circuit design. These are more in tune with hobbyists working on their individual projects, or students learning to work on EDA. But for professional purposes, system based tools remain the better choice.

"An efficient tool should be in a system. This removes dependence on the Internet while working," says Sreekanth Tamanna, country manager for India, Pakistan, Bangladesh and Sri Lanka, Altium Ltd.

Processing time in a Cloud based tool has also increased. If you are working in a closed network, you have to often confirm you have the latest version in your designs. Improved performance, more complex designing, local storage and better working are some advantages with

Altium 16
Designer

